# RescuSphere

## Emergency help in one tap




MOBILE APPLICATION DEVELOPMENT | MINI PROJECT


- **Aarish Bhanwadia** - 12302080601001
- **Dhruv Talati** - 12302080601019
- **Kirtan Chauhan** - 12302080601055

A D Patel Institute of Technology, CVM University

# Getting help takes too many steps

In an emergency, every second matters — but using existing methods is complicated and stressful.

1

### Right helpline

112, 108, 100 or 101?
Hard to recall under stress.

2

### Location

Hard to describe exactly where you are.

3

### Family and friends

Manual calls and texts cost precious time.

4

### Medical information

Responders don't know blood group or allergies.

5

### Describing it

No simple way to state type and severity.

6

### False alarms

A one-touch alert needs a safeguard.

# Existing solutions leave a gap

Current emergency solutions solve only a part of the problem, making them incomplete and stressful to use.

01

**Dial 112 / 108 / 100 / 101**
Call emergency helpline numbers

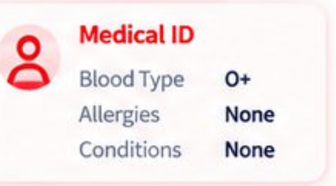
**Needs speech and the right number**
Hard to recall under stress.




02

**Phone Medical ID**
Store basic medical details

**Passive only**
No alert, no GPS, no incident history.




03

**112 India app**
Government emergency app

**Built for dispatch**
Not a personal contact circle
or detailed medical profile.




04

**bSafe, Google Personal Safety**
Safety and location sharing apps

**Need an account, internet and cloud storage**
Not suitable for quick, offline emergency use.



**None combines alerts, medical data, hotlines and history in one offline app.**

# One app for every emergency action

Quick access to help, alerts, and important information — all in one simple, offline-first app.

01

### Dial 112 / 108 / 100 / 101

Call emergency helpline numbers with one tap.

02

### Phone Medical ID

Store and access important medical details (blood group, allergies, conditions).




03

### SOS with a cancel window

A 3-second countdown stops false alarms.

04

### Private by design

SQLite on the device, no cloud. Works offline with core features — no internet required.

**One screen. One tap. No account.**

Essential help, information, and safety — available anytime, anywhere.

# Core features

Essential tools for quick help, safety and information — all in one simple, offline-first app.

01

## SOS Alert

Countdown, GPS and a pre-filled SMS to all contacts.

02

## Contacts and hotlines

One-tap calls to 112, 108, 100, 101 or a contact.

03

## Smart Triage

Type, severity and quick info in three taps.




04

## Medical ID

Blood group, allergies and conditions in each alert.

05

## Incident log

Filter, resolve, reopen, delete, open in maps.

06

## Share and strobe

Share live location or flash the screen red.

# How an SOS alert works

From one tap to real help — a simple, fast and reliable emergency flow.

**1 Tap**

Large SOS button on Home screen.



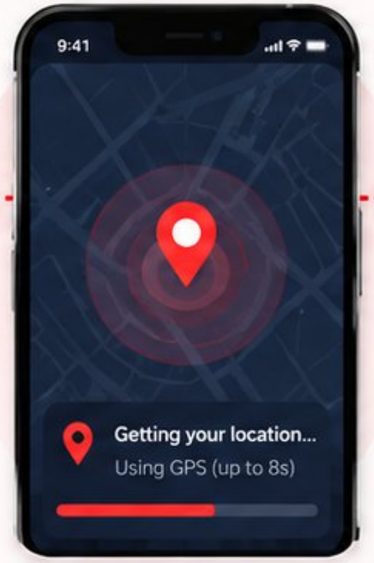

**2 Countdown**

3 seconds, tap again to cancel.



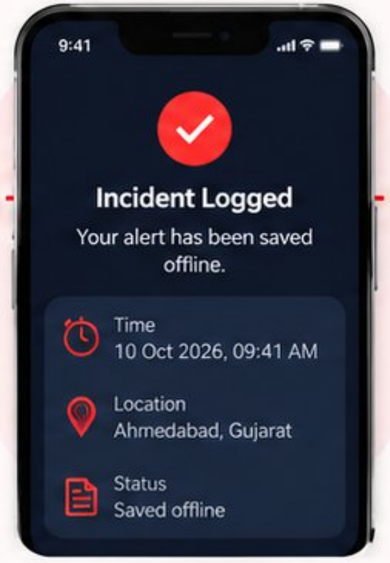

**3 Locate**

GPS in 8 s, last-known fallback.



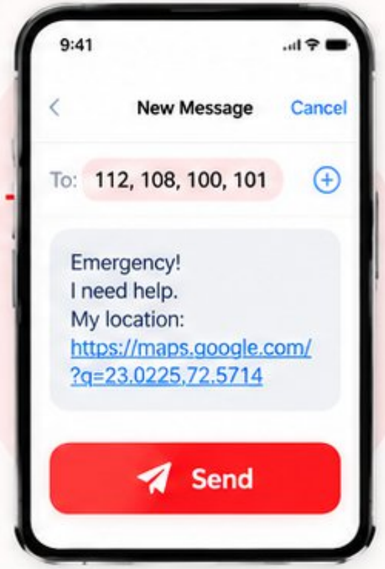

**4 Log**

Active incident is saved.




**5 Alert**

SMS opens; user presses Send.



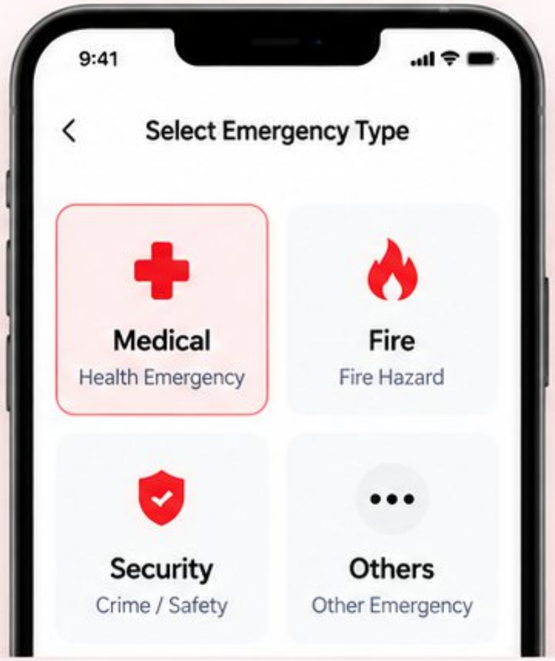

**The incident is logged even if GPS or contacts are missing. No silent background sending.**

# Smart Triage in Three Taps

Get the right help faster by sharing the type, severity and key details of the emergency.

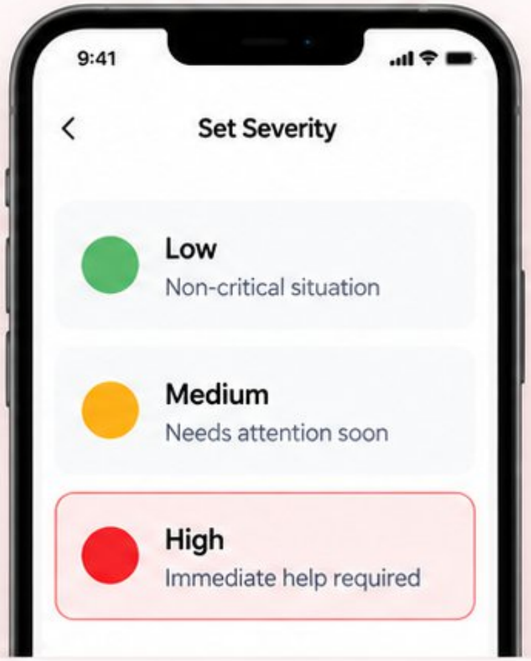
## 01 Choose Type

Select the type of emergency from quick options.



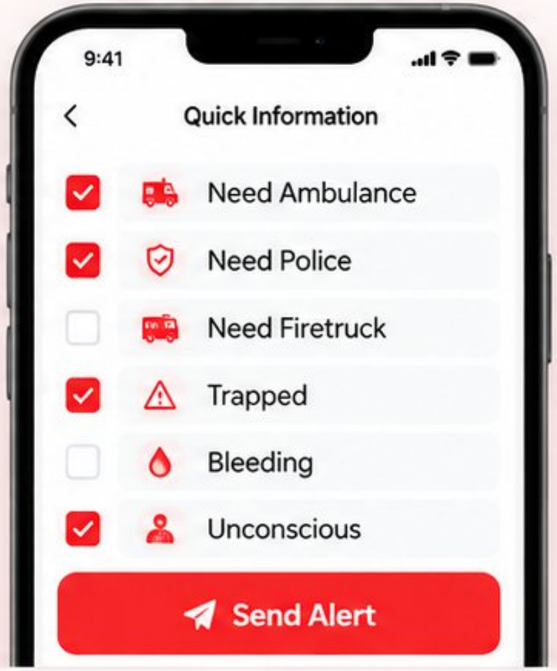

## 02 Set Severity

Choose the severity level to help responders prioritize.




## 03 Add Quick Info

Select key details to share with responders.




**Same alert flow as SOS. Type, severity and quick info appear in the SMS and incident log.**

# Layered architecture



### Presentation Layer
Home, Smart Triage, Medical ID, Incident Logs

UI / Screens

### State Layer
AppProvider holds profile, contacts and logs

State Management

### Services Layer
EmergencyService: GPS, vibration, calls, SMS, maps

Core Functionality

### Data Layer
DatabaseHelper and models over SQLite

Local Database

## Design principles

**Speed**
SOS is the largest control

**Safety**
Countdown, cancel, confirmations

**Privacy**
Local data, no sign-up

**Structure**
Screens go through AppProvider

# Tech stack and local data

## Tech stack

**Flutter / Dart**
One code base, animated UI

**provider**
State management

**sqflite**
On-device SQLite

**geolocator**
GPS and permissions

**url_launcher**
SMS app, dialer, maps




All data stored locally on device (SQLite)

## Database: rescusphere.db

**medical_profile** — User's medical details
name, bloodType, allergies, medicalConditions

**ice_contacts** — Emergency contacts
name, relation, phoneNumber

**incident_logs** — SOS history
timestamp, type, status, summary, latitude, longitude

**Offline by design:** All personal and incident data is stored locally on the device using SQLite. No cloud storage.

# Testing and findings

**16**
Test cases run

**6**
Bugs logged

**0**
Blocking bugs

## Test setup

**Device**
OnePlus Nord CE 2 Lite

**Android version**
Android 14

**Builds tested**
Debug and release APK

**Testing**
Flutter test unit tests




## Priority fixes

**Reword "Alerts sent!"** — UI/UX
User still presses Send

**Validate name and phone number** — Validation
Check before saving

**Remove sample contact** — Data
Before first SOS

# Limitations and future scope

## Current limits

**SMS opens in the phone app**
User has to manually press Send

**Data stays on device**
No cloud backup, stored locally (SQLite)

**Hotline numbers are India-specific**
Pre-configured for Indian emergency numbers

**Needs GPS and mobile network**
Location and SMS require GPS and connectivity




## Future scope

**Background SMS sending**
Send SMS automatically with permission

**Continuous live-location sharing**
Share real-time location until the incident is resolved

**Login and cloud backup of Medical ID**
Sync and backup Medical ID securely

**Nearby hospitals and police stations**
Show nearby emergency services on maps

**Alarm sound, shake or volume triggers**
Additional alert mechanisms for emergencies

# Thank you!

